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A daily rhythm of cell proliferation in a songbird brain

Vladimira Hodova, Valentina Maresova, Rebecca Radic & Lubica Kubikova✉

Neurogenesis is an active process of creating new neurons in the neurogenic zone. It is influenced by many factors, including the circadian system, which is synchronized by light. Neurogenesis in laboratory rodents peaks at night, and the rodents are nocturnal, contrary to humans that are active during the day. Here, we studied whether proliferation and apoptosis exhibit a daily rhythm in the brain of the diurnal songbird zebra finch (*Taeniopygia guttata*) and whether the cell proliferation peaks during the dark phase of the day, as in rodents. We injected the birds with the cell proliferation marker 5-ethynyl-2′-deoxyuridine (EdU; thymidine analog), quantified the number of dividing cells in the neurogenic ventricular zone (VZ), and measured mRNA expression of clock genes as well as genes indicating cell proliferation or apoptosis. First, we confirmed the daily rhythms of the clock genes. Next we found that proliferation along the whole VZ did not exhibit a daily rhythm. However, proliferation in the central ventral part of the VZ, i.e. "the hot-spot" area, showed a daily rhythm of proliferation. The highest number of newborn cells was detected in the dark phase of the day. The relative expression of the apoptotic genes *caspase 3*, *Bcl-2*, and *Bax* as well as the proliferating cell nuclear antigen (*PCNA*) did not show any rhythm. In summary, our results show that cell proliferation in the "hot-spot" region of the VZ in diurnal songbirds shows rhythmic activity over a period of 24 h and that the maximum cell proliferation occurs in the passive phase. This study may have implications for understanding the mechanisms underlying the daily regulation of brain cell proliferation in different species.

Keywords Circadian rhythm, Ventricular zone, EdU, Clock genes, QPCR, Neurogenic

Neurogenesis (the birth of new neurons) is most widespread during embryonic development, but new neurons also arise in the adult brain. While the majority of neurogenesis studies have been performed in rodents[1], songbirds can be a better model at translating discoveries to humans[2].

In mammals, adult-born neurons originate in the subventricular zone of the lateral ventricles and in the subgranular zone of the hippocampal dentate gyrus[3,4]. The main neurogenic niche in songbirds is the ventricular zone (VZ) lining the walls of the lateral ventricle. Newborn neurons disperse from the VZ and differentiate in many areas of the forebrain, including the hippocampus and song control regions HVC lying in the pallium (cortex-like structure) and Area X of the striatum[5–7]. In this respect, songbirds are more similar to humans, in which new neurons migrate to the striatum and not to the olfactory bulb as in rodents[8,9]. Furthermore, song learning shares many parallels with speech learning[10], and both songbirds and humans are diurnal, whereas traditional rodent models are nocturnal. Light drives the activity of songbirds[11], whereas in laboratory rodents, light suppresses physical activity[12,13]. Physical activity stimulates neurogenesis[14–16] and intriguingly, growing evidence suggests that neurogenesis is also modulated by the circadian system, which is synchronized by light[17–20]. For instance, chronic jet lag has a detrimental effect on hippocampal neurogenesis in adult rats[21] and mice[22], and chronic phase shifts reduce hippocampal cell proliferation in hamsters[23].

The circadian system generates circadian rhythms of molecular, physiological, and behavioral processes with an approximately 24-hour period and it is hierarchically organized[18,24]. The light/dark cycle is considered to be the strongest external cue, which is why light is called a time giver (Zeitgeber)[18]. The received signal is sent to pacemakers, which retain circadian rhythms and influence rhythmicity in the periphery. In birds, the pacemakers are the retinae, pineal gland, and as in mammals the suprachiasmatic nuclei of the hypothalamus. The structures are independent oscillators and create a central circadian system in birds[17,25]. The molecular basis of circadian oscillations involves positive (Bmal1 and Clock transcription factors) and negative (period [Per2, Per3] and cryptochrome [Cry1, Cry2]) elements in a transcriptional-translational feedback loop within cells[26–29]. At the beginning of the cycle, BMAL1 and CLOCK heterodimerize and bind to the E-box elements of the *Per* and *Cry* genes, which enhances their transcription. The PER and CRY proteins then bind to the *Bmal1* and *Clock* promoters, thus leading to the inhibition of their transcription. Once the PER and CRY proteins degrade, their inhibitory effects are released, and BMAL1 and CLOCK can re-enter the cycle[27].

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Circadian clock genes can regulate the expression of crucial cell cycle regulators, enabling the synchronization of cell cycle states within a cell population. In mammals, the *Bmal1* and *Per2* genes are essential for maintaining circadian rhythms and can control neurogenesis. The expression of *Per2* is high during the light phase and the cell proliferation rises in the dark phase with the decrease of *Per2*. PER2 is essential for the circadian-regulated cell-cycle entrance, its ablation leads to high levels of cell proliferation, and thus the intact *Per2* prevents cell proliferation in the light phase[30]. BMAL1 has a dual role in restricting both entrance as well as exit from the cell cycle in the brain and in retina[30,31]. The expression of BMAL1 is strongly repressed by nuclear receptors REV-ERBα and REV-ERBβ, which are crucial circadian timekeepers[32]. This repression leads to changes in the expression of factors associated with cell proliferation[33]. On the other hand, the arrhythmic mice lacking the circadian genes *Cry1* and *Cry2* produced normal levels of neuronal cells[19].

Beyond regulating brain cell proliferation in neurogenic zones, circadian clock genes also influence neuronal differentiation[33]. The song control nuclei HVC and Area X contain mRNA for BMAL1 and Per2. BMAL1 shows a daily rhythm in Area X and both, BMAL1 and Per2 have daily rhythms of the expression in HVC[34].

Many of the newborn neurons recruited to HVC do not survive one month[35], and they die via apoptosis[36]. Thus, neurogenesis is closely linked with cell death. In seasonal songbirds, song circuits undergo intense changes during each breeding season and cell death precedes neurogenesis to maintain homeostasis[37]. In a non-seasonal breeder zebra finch (*Taeniopygia guttata*), newborn neurons in the HVC do not replace old neurons, and the density of the HVC neurons increases with age[38]. Nevertheless, neuronal death triggers the recruitment of new neurons to the HVC in male zebra finches[39].

The aim of this study was to determine whether there is a daily rhythm in brain cell proliferation in songbirds and whether the proliferation levels are greater during the light phase when the animals are physically active or in the dark phase with low PER2. We measured the 24-hour mRNA profiles of six core clock genes, the mRNA profiles of the nuclear antigen of proliferating cells (PCNA) and three apoptotic genes, and quantified the proliferating cells in the VZ using a thymidine analog incorporated into the DNA of dividing cells.

Materials and methods

Animals

Thirty-six adult male zebra finches approximately one year old were used. In order to limit the variability within the time-point group, we only used males because sex differences in neuronal recruitment to vocal control regions have been reported[40]. Additionally, the male (but not the female) zebra finch brain contains the vocal regions used as markers of the rostral and caudal positions for cell counting (detailed below). The birds were bred and housed in the animal facilities of the Centre of Biosciences, Slovak Academy of Sciences. They were maintained under the 12L:12D regime. Food and water were available *ad libitum*. All procedures were approved by the State Veterinary and Food Administration of the Slovak Republic and Ethical Committee in accordance with the relevant guidelines and regulations, including the ARRIVE guidelines.

EdU injections and tissue processing

All the birds were weighed and placed in sound-attenuating boxes for 2–4 days. Their songs were recorded via Sound Analysis Pro 2011 software. Only the birds that were singing during this time were included in the study. As higher song sequence variability is associated with higher neurogenesis in a closely related species, Bengalese finch (*Lonchura striata var. domestica*)[41], we wanted to rule out the possible effect of the song on cell proliferation. The birds were evenly divided into six balanced groups ($n = 6$/group) according to the weights and complexity of their songs. The weight of the birds did not differ among the groups ($p = 0.95$, ANOVA, range from 17.28 ± 1.18 to 17.73 ± 1.31). The group always consisted of two birds with the most complex song containing 6–8 different syllables, two birds with a song containing 4–5 different syllables with repetition, and two birds with a simple song containing 3–4 different syllables. After the songs were recorded, the birds were kept in $90 \times 60 \times 43$ cm cages with perches, swings, and a water bath, with auditory and visual contact. Two hours before sacrifice, they were injected intramuscularly with the exogenous marker of cell proliferation, the thymidine analog 5-ethynyl-2´-deoxyuridine (EdU; Carl Roth, Karlsruhe, Germany; dose 41 mg/kg, concentration 10 mg/ml). The 2 h period leads to a sufficient number of labeled cells[41,42]. After EdU administration, the birds were sacrificed at the times ZT2, ZT6, ZT10, ZT14, ZT18, ZT22 according to the group assignment during the 24-hour period (Fig. 1A). ZT0 was considered as the beginning of the light phase and it was at 7 AM. The birds from the dark phase of the day were injected and sacrificed with a black cap on their heads to minimize light exposure. The brains were quickly removed from the skull, frozen in Tissue-Tek OCT Compound (Sakura, Japan), and stored at −20 °C.

The brains were cut coronally using a Leica 1800 cryocut. Section 10 µm thick were taken every 240 µm (Fig. 1B), mounted on silanated slides, and stored at −20 °C. The brains were also cut into 60 µm sections, containing the central part of the VZ and hypothalamus. These sections were placed into microeppendorf tubes and stored at −80 °C until mRNA isolation. There were 8 sections in each tube.

EdU-positive (EdU+) cell labeling and analysis

The sections on the slides were fixed with paraformaldehyde (PFA) for 5 min and washed 3 times for 5 min each in PBS (pH 7.4). Nonspecific binding was blocked for 1 h in blocking solution (PBS containing 0.1% bovine serum albumin and 0.3% Triton X-100). The sections were washed 3 times for 5 min each in PBS and incubated in an EdU Click-IT Cocktail (containing deionized H_2O, reaction buffer, catalytic solution, 6-FAM azide, and buffer additive) for 1 h in the dark. The commercially available ClickTech EdU Cell Proliferation Kit 488 (Carl Roth, Karlsruhe, Germany) was used for EdU labeling. Then, the sections were washed again 3 times for 5 min each in PBS, rinsed in deionized water, and coverslipped with the Vectashield antifade mounting medium with DAPI (Vector Laboratories, Burlingame, USA).

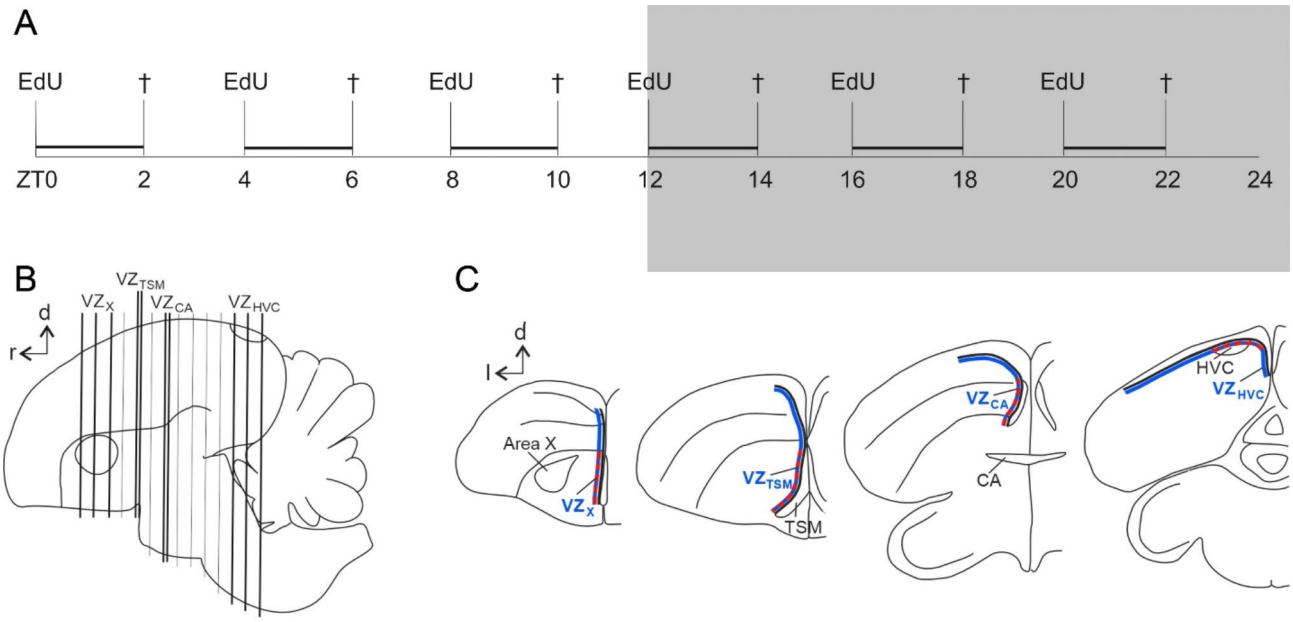

Fig. 1. Experimental setup. (**A**) Schematic timeline of EdU administration and sacrifice (cross symbol) during the 24-hour period. The horizontal thick line connects the time point of the EdU administration and the sacrifice of the same group of birds. Each group (ZT2, ZT6, ZT10, ZT14, ZT18, and ZT22) consisted of six individuals. The shadow area represents the dark phase during the 12L:12D cycle. (**B**) Schematic drawing of the cutting (vertical lines) on the sagittal section showing the levels of the ventricular zone. The thick vertical lines highlight the sections with the VZ containing the vocal nucleus Area X (VZ_X), *tractus septomesencephalicus* (VZ_{TSM}), *commisura anterior* (VZ_{CA}), and vocal nucleus HVC (VZ_{HVC}). (**C**) Schematic drawing of the quantification of the EdU^+ cells on the coronal section at four anatomical levels in the whole ventricular zone (blue line) and in the ventral part of the ventricular zone (red line). *ZT* zeitgeber, *VZ* ventricular zone, *d* dorsal, *r* rostral, *l* lateral.

The images were taken using the Leica Microsystems software LAS AF6000 connected to a Leica DM5500 fluorescent microscope and a Leica DFC 340 FX camera. A 10x lens was used. Quantification was performed using PHOTOSHOP CS 6 software. The number of EdU^+cells was quantified at four anatomical levels from the rostral to the caudal part of the VZ, as described previously[41]. These sections included the vocal nucleus Area X (VZ_X), *tractus septomesencephalicus* (VZ_{TSM}), *commisura anterior* (VZ_{CA}), and vocal nucleus HVC (VZ_{HVC}, Fig. 1C). For each bird and each level, 3–4 sections were calculated. The EdU^+ cells were counted along the ventricle in the whole section (Fig. 1C, blue line). The values were recalculated as the number of cells per millimeter of length of VZ and the numbers from 2 hemispheres on every level of the VZ were averaged per bird. We also quantified the number of EdU^+ cells in the ventral part of the VZ separately (Fig. 1C, red line) because it includes the "hot-spot" area, i.e. the area with the highest degree of cell proliferation[42,43]. The degree of cell proliferation in this "hot-spot" region was calculated as the average number of EdU^+ cells at the VZ_{TSM} and VZ_{CA}.

Gene expression

The 60 µm thick brain sections were homogenized in 1 ml of TRI reagent (Molecular Research Center, Cincinnati, USA) in microeppendorf tubes using an ULTRA TURRAX® IKA® T18 basic homogenizer (10,000 rpm, IKA - Labortechnik, Germany). The phenol-chloroform extraction method was used to isolate the mRNA. Throughout the process, isopropanol was used for the precipitation and 75% ethanol was used for the purification of the mRNA. The mRNA was resuspended in RNase-free water. The concentration and purity of the isolated mRNA were verified using a Multiskan GO spectrophotometer (Thermo Fischer Scientific, Valtham, MA, USA) at 260/280 and 260/230 nm ratios of absorbance. The mRNA was considered pure when the value of the 260/280 nm absorbance was close to 2. Then, the mRNA was treated with DNAse I before reverse transcription (RapidOut DNA Removal Kit, Thermo Fischer Scientific, Valtham, MA, USA). cDNA was synthesized with the RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, Valtham, MA, USA).

The analysis of gene expression was performed on a LightCycler® Nano Real-Time PCR Instrument (Roche Applied Science, Penzberg, Germany), using Maxima SYBR Green/ROX qPCR Master Mix (2x) (Thermo Fisher Scientific, Valtham, MA, USA), with a final reaction volume of 20 µl. We measured the gene expression of clock genes (*Per2*, *Per3*, *Cry1*, *Cry2*, *Bmal1*, *Clock*), marker of proliferating cells (PCNA), and apoptotic genes (*caspase 3*, *Bax*, *Bcl-2*). We used the ribosomal protein L13 (RPL13) as a reference gene. The primers were designed with Primer-BLAST using a sequence of nucleotides found in GenBank. The primers used were at a concentration of 5 µM, and their sequences are shown in Table 1.

Gene	Accession number	Forward primer (5′-> 3′)	Reverse primer (5′-> 3′)	Product length (bp)
Per2	XM_012575458.1	ACAGCCATTCCAGTACAGCC	GCGGTTTTCGTGGTTTCCTC	189
Per3	XM_012570379.4	CCGAAAAAGCCAACAGCGAA	GCACTGATGGTACTGGGCTT	130
Cry1	XM_030263132.3	TCAGTCCCTACCTCCGCTTT	TTGTTAGTCGCCGCTGTGTA	157
Cry2	XM_030274299.3	AGTCCCTACCTGCGTTTTGG	GAAGAACTCCCGCCACAGAA	135
Bmal1	XM_021527552.1	CTTCGAGGTGCTACAAACCCA	AATCCATTTGGTGCCCTGAG	101
Clock	XM_041715750.1	GTCTGGAATGACTCAGCCTGT	AACTCTTCCTGCTGCCGATG	126
PCNA	NM_001245444.1	AAAATCCTGAAATGCGCGGG	ACTTTCTCCTGGTTGGGTGC	104
caspase 3	XM_030271265.3	TGTCACCCCGTTCAGGTACA	TGCTTCGCTTGCTGTGATCT	156
Bax	XM_041712988.1	GGGCTATTGTCCCACTTCGG	GGGGTAGTGGGGGACAAAAT	135
Bcl-2	XM_041712988.1	GATGTGTGTGGAGAGCGTCA	TTTTACGCTGCCCTCTCTGT	140
RPL13	XM_002195737.6	CGGAACAAGTCCACCGAGTC	GTCTCCCTTCTTGGGCATGG	108

Table 1. Sequences of the primers used for real-time PCR.

The RT-PCR program consisted of an initial denaturation step for 10 min at 95 °C, an amplification step containing 42 cycles of 15 s at 95 °C, 20 s at 55 °C, and 20 s at 72 °C, followed by a hold step for 10 min at 95 °C and then a melting curve step of 60 s at 60 °C and 1 s at 97 °C. The data were analyzed with LightCycler® Nano Software 1.1. The relative gene expression was calculated using a formula (efficiency of target gene^$-Cq_{target\ gene}$/efficiency of reference gene^$-Cq_{reference\ gene}$). Before the calculation, we always tested the rhythmicity of the reference gene. The final values for each gene were expressed as a ratio of individual values of relative expression to the average of all values of relative expression.

Statistical analyses

All the data were analyzed using the freely available cosinor analysis Cosinor.Online[44]. In addition to the p value, we also reported an acrophase (the point at which the rhythm reached its maximum from the beginning of the light phase) and amplitude (one half of the difference between the peak and trough of the rhythm). The graphs were created in GraphPad Prism 6 using a cosinor analysis. To compare the time-point groups with the highest and lowest number of proliferating cells in a given area, t test was used. Lateralization of cell proliferation was examined using paired t-test. ANOVA was used to compare the weights of the birds in the experimental groups.

Results

Clock gene expression

First, we analyzed whether the core clock genes exhibited the daily rhythms to verify that the birds were synchronized to the LD cycle. The cosinor analysis revealed significant daily rhythms in the expression of the clock genes *Per2* ($p = 0.0003$, $F_{2,33} = 10.22$, $r^2 = 0.38$), *Per3* ($p < 0.00001$, $F_{2,33} = 48.58$, $r^2 = 0.75$), *Cry1* ($p = 0.0015$, $F_{2,33} = 7.93$, $r^2 = 0.32$), and *Cry2* ($p = 0.0104$, $F_{2,33} = 5.26$, $r^2 = 0.24$), with maximum transcription at the beginning of the light phase (acrophase for *Per2*: 3.00; *Cry2*: 3.51; Fig. 2A, D), in the middle of the light phase (acrophase for *Cry1*: 7.10; Fig. 2C) or at the end of the dark phase (acrophase for *Per3*: 23.93; Fig. 2B). The expression of *Bmal1* gene also exhibited a significant daily rhythm ($p < 0.00001$, $F_{2,33} = 60.92$, $r^2 = 0.79$; Fig. 2E) and was in antiphase to the *Per* and *Cry* genes, with an acrophase of 11.15. The expression of *Clock* was stable during the 24-hour period ($p = 0.38$, $F_{2,33} = 1.01$; Fig. 2F). Thus, we confirmed the daily rhythms of the clock genes in our samples.

Proliferation at the DNA level

Proliferation within the whole VZ

The cosinor analysis did not reveal any significant daily rhythm in the number of EdU^+ cells per mm length counted in the whole VZ at four anatomical levels (Fig. 1C, blue line) at any level (VZ_X: $p = 0.17$, $F_{2,33} = 1.90$; VZ_{TSM}: $p = 0.23$, $F_{2,33} = 1.51$; VZ_{CA}: $p = 0.58$, $F_{2,33} = 0.55$; VZ_{HVC}: $p = 0.93$, $F_{2,33} = 0.07$; Fig. 3A-D, blue symbols) or when we averaged the number of cells at all anatomical levels of the VZ ($p = 0.26$, $F_{2,33} = 1.42$, not shown). We then calculated the total number of new cells (not shown) and we summed the number of cells at the four anatomical levels of the VZ (not normalized per mm length). We found that the total number of proliferating cells at these levels or in the whole VZ did not exhibit any daily rhythm either ($p > 0.2$). Furthermore, we examined whether there was lateralization in the number of EdU^+ cells and found that there was no difference between the hemispheres ($p > 0.2$).

Proliferation in the "hot-spot" region of the VZ

By visual examination, we noticed considerable differences in proliferation in the rostrocaudal as well as dorsoventral directions, which is consistent with the previously published data[42,43], and the highest new cell density was in the ventral central part of the VZ. Therefore, we also examined proliferation specifically in this "hot-spot" area (Fig. 1C, red line at the VZ_{TSM} and VZ_{CA} levels). Cosinor analysis revealed that the number of EdU^+ cells in the "hot-spot" area exhibited the daily rhythm (VZ_{TSM}: $p = 0.015$, $F_{2,33} = 4.74$, $r^2 = 0.22$, Fig. 3B red line and symbols, Fig. 4A; VZ_{CA}: $p = 0.042$, $F_{2,33} = 3.48$, $r^2 = 0.17$, Fig. 3C red line and symbols, Fig. 4B). The maximum proliferation occurred in the dark phase of the day around ZT15 in both VZ_{TSM} (acrophase = 15.05) and VZ_{CA} (acrophase = 14.91) and amplitude of the rhythm was 1.99 for VZ_{TSM} and 2.75 for VZ_{CA}. When the troughs and peaks of the rhythms were compared (ZT2 vs. ZT14), a t-test revealed significant differences in the

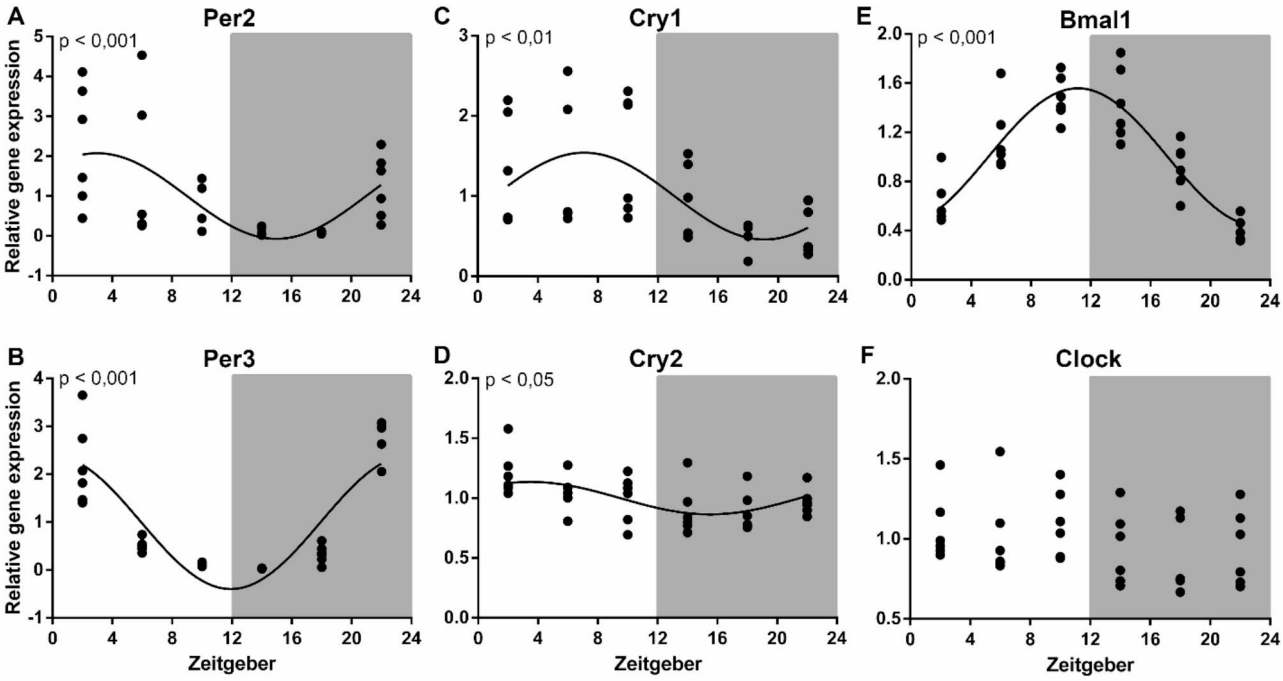

Fig. 2. Relative gene expression of clock genes during the 24-hour period. Gene expression of (**A**) *Per2*, (**B**) *Per3*, (**C**) *Cry1*, (**D**) *Cry2*, (**E**) *Bmal1*, and (**F**) *Clock*. Each group (time points ZT2, ZT6, ZT10, ZT14, ZT18, and ZT22) consisted of $n = 6$ individuals. Cosinor analysis was used to determine significant differences.

number of EdU$^+$ cells in both the VZ_{TSM} ($p = 0.02$) and VZ_{CA} ($p = 0.01$). Generally, we observed a significant daily rhythm of counted EdU$^+$ cells in the whole "hot-spot" region of the VZ ($p = 0.007$, $F_{2,33} = 5.73$, $r^2 = 0.24$, Fig. 3E, red line and symbols) with maximum proliferation in the dark phase of the day (acrophase = 14.97) and an amplitude of the rhythm 2.37. There was a significant difference between the time-points with the lowest and highest proliferation (ZT2 vs. ZT14) in the whole "hot-spot" region of the VZ ($p = 0.003$; t-test). We wanted to know whether there might be any effect of the song type on the cell proliferation. We plotted the same data as in Fig. 3E, distinguished birds assigned to the three song groups (song containing 6–8, 4–5, or 3–4 different syllables; Supplementary Fig. 1) but we did not see any relationship with the song type and the number of cells. Finally, we wanted to determine whether brain cell proliferation exhibits a rhythm only in the ventral central part of the VZ or also in the ventral rostral and caudal parts of the VZ. We also examined the numbers of EdU$^+$ cells in the sections from VZ_X and VZ_{HVC} (Fig. 1C, red line on the levels VZ_X and VZ_{HVC}). We found that the number of EdU$^+$ cells in these regions did not exhibit a daily rhythm (VZ_X: $p = 0.44$, $F_{2,33} = 0.83$; VZ_{HVC}: $p = 0.72$, $F_{2,33} = 0.33$; Fig. 3A, D, red symbols; Fig. 4C for VZ_X).

Daily rhythm in PCNA mRNA expression

Next, we attempted to determine whether there is a daily rhythm of proliferation also at the mRNA level. We used qPCR and determined the expression of the cell proliferation marker, the nuclear antigen of proliferating cells (PCNA). Cosinor analysis revealed that there was no significant difference in the daily rhythm of PCNA ($p = 0.63$, $F_{2,33} = 0.48$, Fig. 5A).

Gene expression of apoptotic genes

Since we revealed that cell proliferation in the "hot-spot" region of the VZ exhibits a daily rhythm, we wanted to know whether a daily rhythm also exists for cell death. The process of apoptosis is regulated by the gene expression of the executioner *caspase 3*, the pro-apoptotic gene *Bax*, and the anti-apoptotic gene *Bcl-2*. In our samples, *caspase 3*, *Bax*, and *Bcl-2* gene expression did not exhibit a rhythm during the 24-hour period ($p = 0.39$, $F_{2,33} = 0.98$; $p = 0.35$, $F_{2,33} = 1.08$; and $p = 0.30$, $F_{2,33} = 1.24$, respectively, Fig. 5B-D). One outlier from timepoint ZT14 was excluded from the analysis of *Bax* gene expression because its value was more than 5 SD away from the average.

Discussion

Neurogenesis is affected by many processes and factors[45]. Here we investigated whether there is a daily rhythm in cell proliferation in the songbird brain. We observed rhythmic activity within a 24-hour period, and the most intensive new cell production occurred at approximately ZT15 in the 12L:12D regime.

In nature, the light-dark cycle entrains the oscillations of the main circadian genes. The circadian clock functions to synchronize behavioral and physiological processes with the natural photoperiod[24]. In this study, we first confirmed that the circadian rhythm of clock genes was not disrupted. The *Per* and *Cry* genes are light-

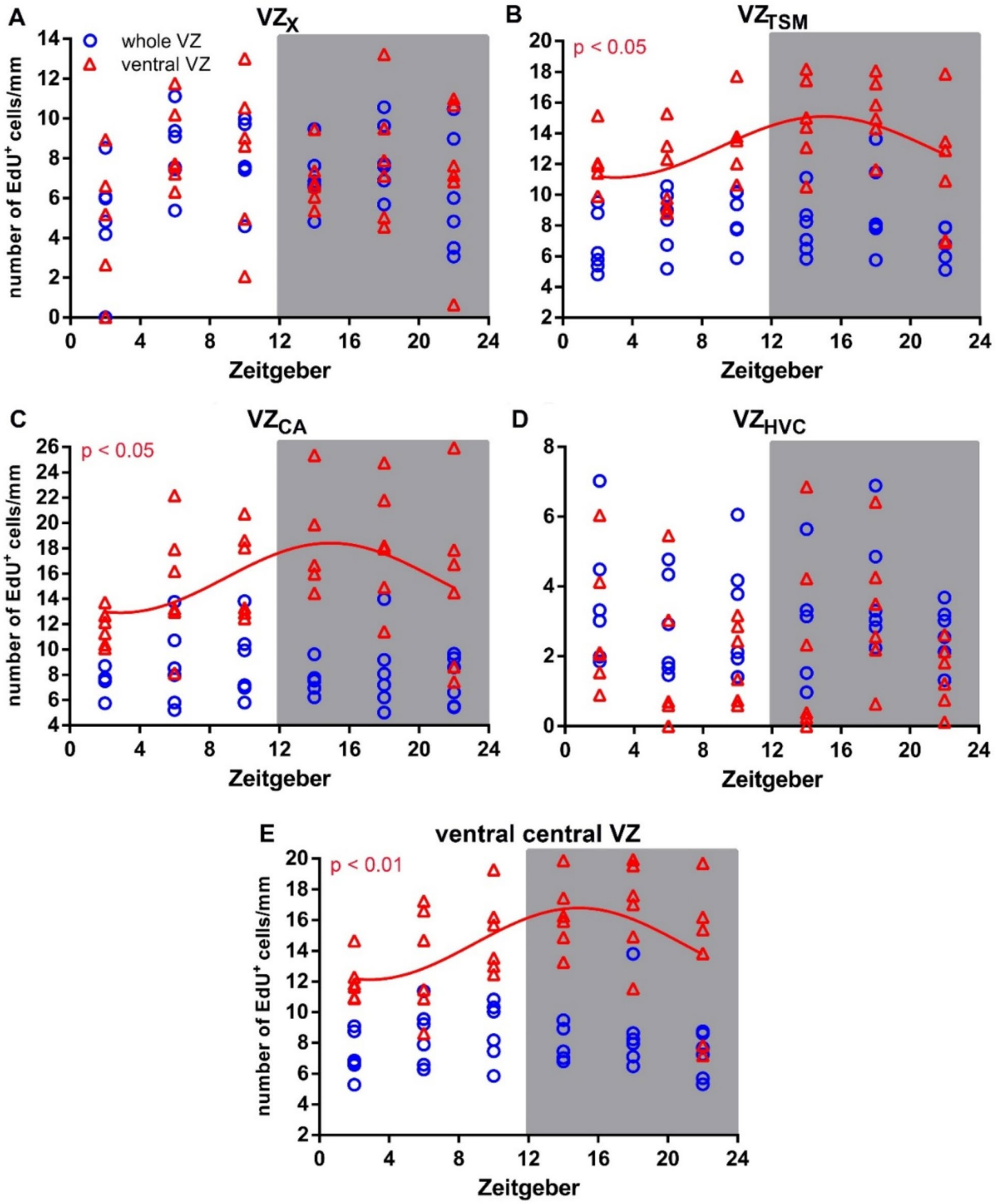

Fig. 3. Proliferation during the 24-hour period in the whole VZ (blue circles) and in the ventral part of the VZ (red triangles and lines). EdU^+ cells were quantified at four levels of the ventricular zone containing: (**A**) the vocal nucleus Area X (VZ_X), (**B**) *tractus septomesencephalicus* (VZ_{TSM}), (**C**) *commisura anterior* (VZ_{CA}), and (**D**) vocal nucleus HVC (VZ_{HVC}). (**E**) ventral central VZ. Each group (time points ZT2, ZT6, ZT10, ZT14, ZT18, and ZT22) consisted of $n = 6$ individuals. Cosinor analysis was used to determine significant differences.

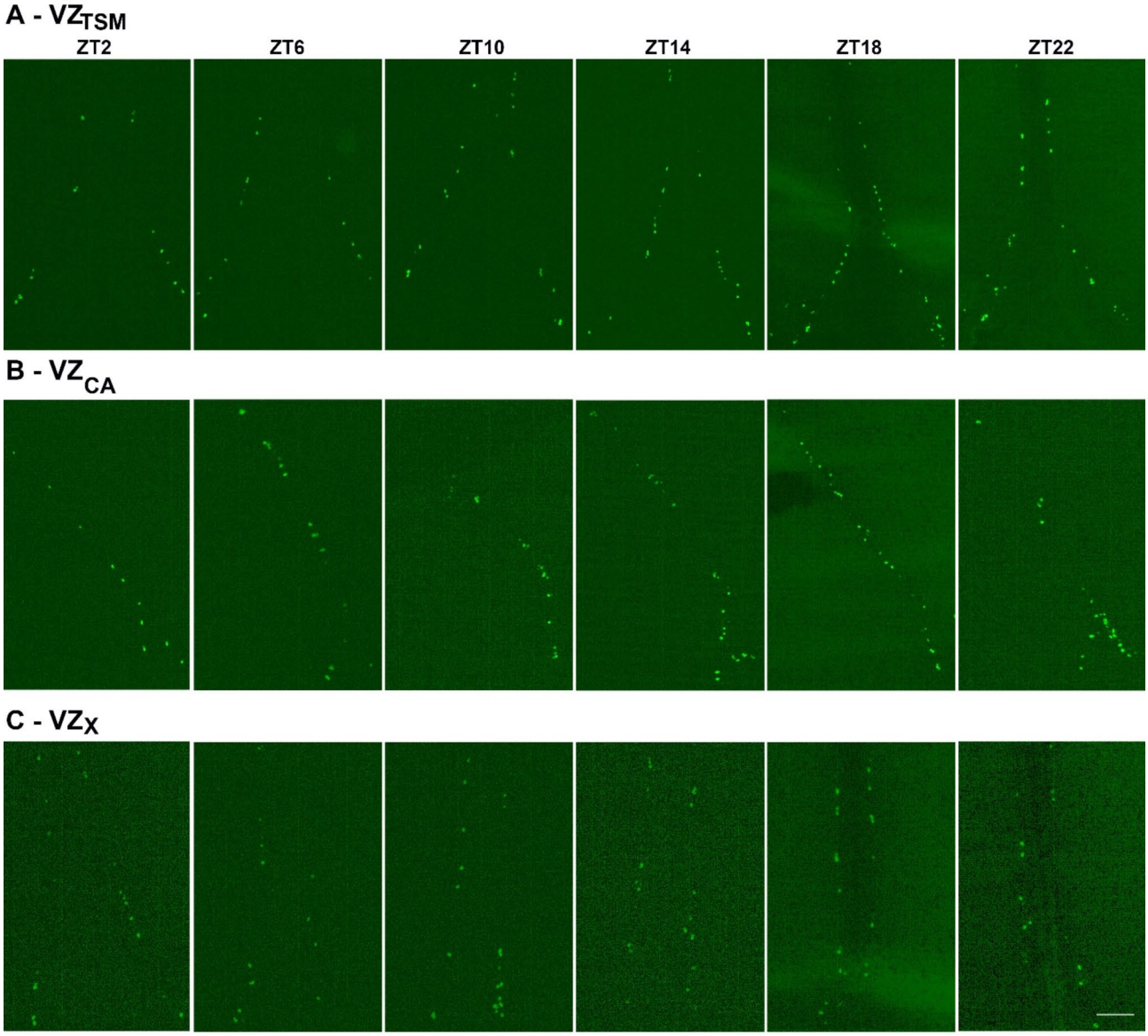

Fig. 4. Photomicrographs of EdU$^+$ cells in the "hot-spot" region of VZ during the 24-hour period. Images showing the ventral part of the ventricular zone containing (**A**) *tractus septomesencephalicus* (VZ_{TSM}), (**B**) *commisura anterior* (VZ_{CA}), and (**C**) Area X (VZ_X). EdU$^+$ cells are the bright dots. The VZ in both hemispheres is shown in A and C, and one hemisphere is shown in B. The scale bar represents 100 µm.

sensitive, the transcription factor BMAL1 is in antiphase to *Per* and *Cry* genes and peaks around the transition from the light to the dark phase, and the transcriptional factor CLOCK is usually arrhythmic during the 24-hour period in the master oscillator in the mammalian hypothalamic suprachiasmatic nuclei[46–49]. Correspondingly, we detected significant daily rhythms in the mRNA expression of the *Per2*, *Cry1*, and *Cry2* genes, with maximum transcription occurring in the light phase of the day. The *Per3* gene also exhibited a significant daily rhythm but peaked at the end of the dark phase. *Bmal1* gene expression exhibited significant rhythm and was in antiphase to *Per* and *Cry* gene expression, with a peak at the transition from the light phase to the dark phase. These results from the whole brain sections closely resemble the clock gene expression reported in previous studies in which the whole-brain homogenates of zebra finches[50] and house sparrows[51], hypothalamus homogenates of spotted munia[52] and readhead buntings[53], and the chicken pineal gland[28,54] were used. *Clock* gene expression did not exhibit a significant rhythm and was stable during the 24-hour period as expected[49,50,54] although in some studies it has been observed to be rhythmic[51,53]. We validated that our experimental setup and methodology allow observing the rhythmic activity of the studied clock genes over 24 h and are suitable for studying daily rhythmicity.

The new cells from the VZ can give rise to glia and various types of neurons within the song system. The new neurons within the striatal Area X are medium spiny neurons[6,55,56] and they arise in the VZ adjacent to the striatum[6,56]. The new neurons in HVC are the projecting neurons and possibly also interneurons[6,57], and

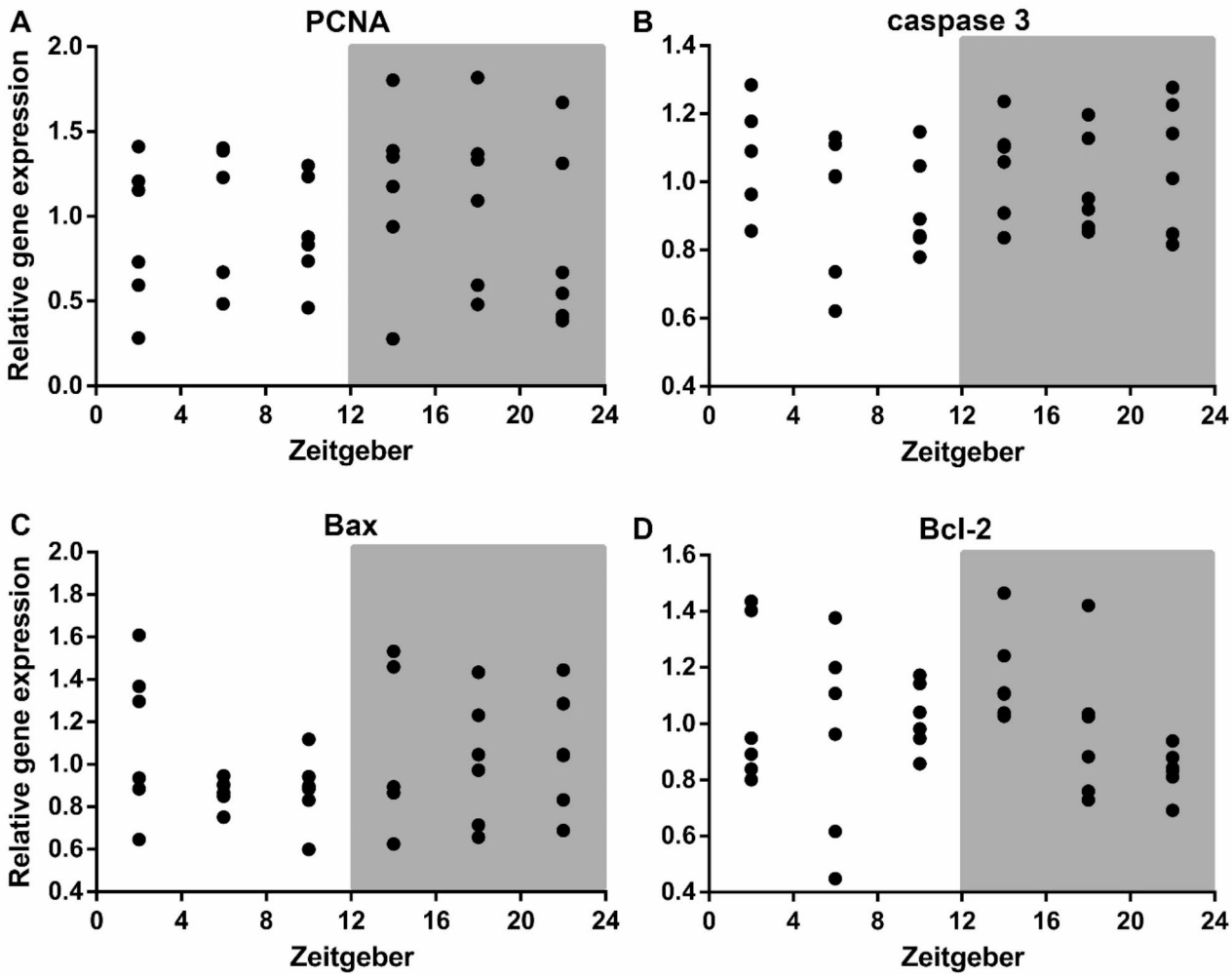

Fig. 5. Relative gene expression of proliferation marker and apoptotic genes during the 24-hour period. Gene expression of (**A**) *PCNA*, (**B**) *caspase 3*, (**C**) *Bax*, (**D**) *Bcl-2*. Each group (time points ZT2, ZT6, ZT10, ZT14, ZT18, and ZT22) consisted of $n = 6$ individuals. Cosinor analysis was used to determine significant differences.

they arise predominantly from the part of VZ above HVC[6]. Here, we studied whether the number of newborn cells changed over 24 h in the adult songbird brain, which, if confirmed, could suggest the temporal control of neural progenitors. When the number of new cells throughout the entire length of the VZ was calculated, we did not find significant changes in cell proliferation. However, the distribution of proliferating cells within the VZ is not uniform and they are denser in the ventral central part of the VZ[42,43]. Therefore, we selected and quantified the number of EdU^+ cells in this region and found that cell proliferation exhibits daily rhythmicity. In adult mice, Kochman et al.[58] did not find daily variation in cell proliferation in the subgranular zone (SGZ) of the hippocampus, but they reported more dividing cells during the light phase in the hilus, which may give rise to glia[59]. Later studies in mice revealed circadian rhythm in cell proliferation in the SGZ, with increased proliferative activity during the dark phase[30,60]. Similarly, we found that proliferation in zebra finch was greater during the dark phase. Since ultraviolet light can induce DNA damage and lead to the accumulation of mutations in a single-celled alga[61], it is possible that there has been selection to shift DNA replication to the part of the day when this radiation is lowest[62]. Perhaps similar processes are at play in zebra finch. Contrary to our results, exposure to artificial light at night has been shown to increase cell proliferation in the VZ and new neuronal recruitment in song control regions, although the intensity of the light in those experiments was very low in comparison to that of daylight, and proliferation was calculated in the whole VZ[63,64]. On the other hand, dim light in Indian house crows (*Corvus splendens*) reduced the number of new young neurons suggesting the negative effect of the light exposure on neurogenesis[65].

Greater proliferation during the night in the rodent hippocampus is under the critical control of proteins in neural progenitor cells, including clock genes and melatonin[66]. The lack of *Per2* abolishes the gating of the entrance to the cell cycle while the ablation of *Bmal1* delays the cell cycle exit and leads to the constitutively high level of proliferation[30] and *Bmal1* affects also the neuronal fate commitment[19]. During the night, the progenitors enter the M-phase of the cell cycle and thus give rise to more progeny[30,60]. Manipulation of circadian clock genes in songbirds has not been done. We did not perform this study under continuous light conditions and

thus our results do not provide evidence of the circadian control of cell proliferation in songbirds. However, the daily expression of the *Per2* and *Bmal1* genes that we detected in zebra finch is consistent with the activity in rodents[48,49]; thus, it can be assumed that the higher cell proliferation at night observed in zebra finch can be controlled by clock genes in a manner similar to that in rodents.

Furthermore, physical activity also increases cell proliferation and neurogenesis in laboratory rodents that are active during the dark phase of their circadian cycle[67]. The increase can be caused by various factors including metabolic factors, muscle-derived myokines or microRNAs which stimulate the production of neurotrophins[68,69]. Zebra finches, similar to humans, are active during the light phase. Exercise in humans causes anti-inflammatory response and increases levels of trophic factors[69] that promote neurogenesis[70]. Zebra finches in our experiment could fly and interact during the day. Kim et al. reported that zebra finches sing almost exclusively during the light phase of the day[71]. Singing induces expression of the brain-derived neurotrophic factor in the song control region HVC, increases neuronal survival[72], and the amount of singing positively correlates with the number of new neurons in the HVC[73]. However, it is not known whether singing also affects cell proliferation. On the other hand, we cannot exclude negative effects on proliferation, such as social stress, that would suppress it during the day.

When investigating transcripts involved in the cell proliferation at the mRNA level, we found that the relative gene expression of *PCNA* did not exhibit any daily rhythm although it is under the strict control of the PER2-mediated circadian clock system and PER2 restricts *PCNA* transcription[74]. One reason may be that it was measured in whole-brain sections, which are large areas. Analysis in a specific part of the brain could reveal more precise results. The second reason PCNA did not exhibit rhythmic activity may be its duration of expression. The half-life of PCNA is approximately twenty hours for cells that turn from cycling to a quiescent state, eight hours for cells that go through the cell cycle, and a certain amount of nondegraded protein can remain in cells after completing the cell cycle[75–77]. Although PCNA immunoreactivity is strongest around the walls of the lateral ventricle in zebra finch, it is broadly distributed throughout the avian brain[40,78]. An alternative to PCNA as a marker of cycling stem cells or progenitor cells at the mRNA level could be Ki-67[79] which is a nuclear protein present in all active phases of the cell cycle[80]. It has been widely used as a marker of proliferating cells in mammals[81,82]. Although immunoreactive proliferating cells expressing Ki-67 have been identified in birds[83], the nucleotide sequence in avian species has not been published. We attempted to examine *Ki-67* gene expression using several sets of mouse and ovine primers (data not shown); however, without success. We hypothesize that the primers had low sensitivity and specificity. Knowing the sequence in zebra finch or at least in other avian species would help to resolve this problem in the future.

Apoptosis as a form of programmed cell death is essential for brain development, homeostasis and plasticity, and the song control system profoundly affects both cell death and neurogenesis annually during each breeding season[37]. Apoptosis is causally linked with neurogenesis; caspase-mediated apoptosis in HVC increases neural stem-cell proliferation in the adjacent VZ during the switch from breeding to non-breeding conditions in male white-crowned sparrows[84], and cell death also induces neurogenesis in zebra finches[39]. Since we observed the daily rhythm of cell proliferation, we were interested in whether daily changes in the expression of apoptotic genes also occur. The overexpression of circadian genes can have antiproliferative effects and lead to apoptosis[85–87]. In our study, we wanted to determine whether cell death has a daily rhythm similar to cell proliferation in the "hot-spot" region of the VZ. We detected apoptosis by analyzing the gene expression of executioner *caspase 3* and the antagonistically working proteins pro-apoptotic *Bax* and anti-apoptotic *Bcl-2*. We did not observe a daily rhythm of *caspase 3*, *Bax*, *or Bcl-2* gene expression in zebra finch males. The reasons explaining our results may be that we measured the expression in the whole sections as oppose to the dissected VZ and also that we measured uncleaved nonactive procaspase 3. Another reason may be its presence in the uncleaved state in cells that already exist. Proteins BCL-2 and BAX have been shown to vary in a circadian manner in mouse bone marrow and their relative abundances are in antiphase to each other: BCL-2 has the maximum peak in the light phase of the day, whereas BAX has the greatest peak in the dark phase of the day[88].

Conclusions

This is the first study of brain cell proliferation in connection with the circadian system in songbirds. We found that the proliferation in the central ventral "hot spot" region of the ventricular zone appears in a daily rhythm and that maximum cell proliferation occurs at the beginning of the dark phase.

Data availability

Data will be made available on request from the corresponding author.

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Author contributions
V.H. and L.K. conceptualized and designed the study, V.H. and L.K. completed the experiments, V.H., R.R., V.M. and L.K. analyzed the data, V.H. wrote the first draft, all authors revised the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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